Robust Waste Material Classification

Do CNNs Learn Materials or Visual Shortcuts?

Task

Classify cardboard, glass, metal, and plastic waste images.

Model

EfficientNet-B0 transfer learning using ImageNet pretrained weights.

Focus

Accuracy + failure analysis under real-world image changes & perturbations.

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Problem and Motivation

Waste sorting errors reduce recycling efficiency and increase contamination.
Material classes can look visually similar in normal photos.
Glass and plastic can both be transparent; metal and plastic can both be shiny.
Main question: does the CNN learn material identity, or visual shortcuts?

Why this is a CV problem

The correct answer depends on texture, reflection, transparency, shape, lighting, and background.

Why accuracy alone is not enough

We also need confusion matrices, robustness testing, and misclassified examples.

Non-CS explanation

Given a waste photo, predict the material and explain when the prediction is reliable.

Dataset Curation and Model Pipeline

Real dataset, controlled split, four training setups, and a key preprocessing insight

Dataset

- RealWaste 4-class subset
- Classes: cardboard, glass, metal, plastic
- Real-world waste photos

Controlled Split

- 70 train / 15 val / 15 test per class
- Synthetic images: **training set only**
- Val/Test remain **real images only**

What is Original / Special

- **Original framing** of public dataset
- Robustness-focused evaluation
- **Normalization OFF** improved results
- Synthetic images were generated only from training images. Validation/test stay real.

Pipeline / Workflow

Data Curation

- Select & clean 4 RealWaste classes
- Balanced train/val/test split
- Synthetic training image generation

Model Pipeline

Loaded EfficientNet-B0 with ImageNet pretrained weights

→ replaced the final layer with a 4-class classifier head
→ trained 4 setups: Baseline, Augmentation, Synthetic, Synthetic + Augmentation
→ compared accuracy, loss curves, and confusion matrices
→ tuned preprocessing and found normalization OFF worked better
→ retrained best setup
→ ran final test + robustness evaluation

Model & Experimental Strategy

MODEL ARCHITECTURE

EfficientNet-B0

Pretrained on ImageNet. Frozen backbone with a custom classifier head.

Why EfficientNet-B0?

Lightweight and efficient, optimized for small-to-medium classification tasks.

Why Transfer Learning?

Small datasets benefit from pre-learned features like edges, textures, and shapes.

EXPERIMENTAL TRAINING SETUPS

1. Baseline
Real training images only

2. Augmentation
Standard techniques: Crop, flip, rotation, color jitter

3. Synthetic
Training on perturbed images only

4. Synthetic + Augmented
Combined approach for maximum diversity

Synthetic Perturbations

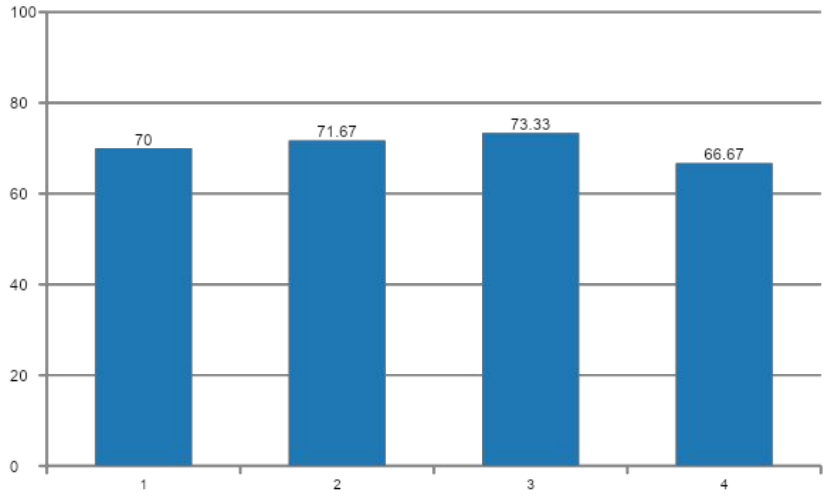
- Blur & Occlusion
- Brightness/Contrast shifts
- Geometric Rotation

Simulation Purpose

Mimics real-world challenges: motion blur, imperfect lighting, partial blockage, and framing inconsistencies.

Main Results: Synthetic Data Performed Best

Accuracy is useful, but the confusion matrices explain where the model still fails.



Best model

Synthetic data: 73.33% test accuracy

Baseline

70.00% test accuracy

Augmentation

71.67% test accuracy

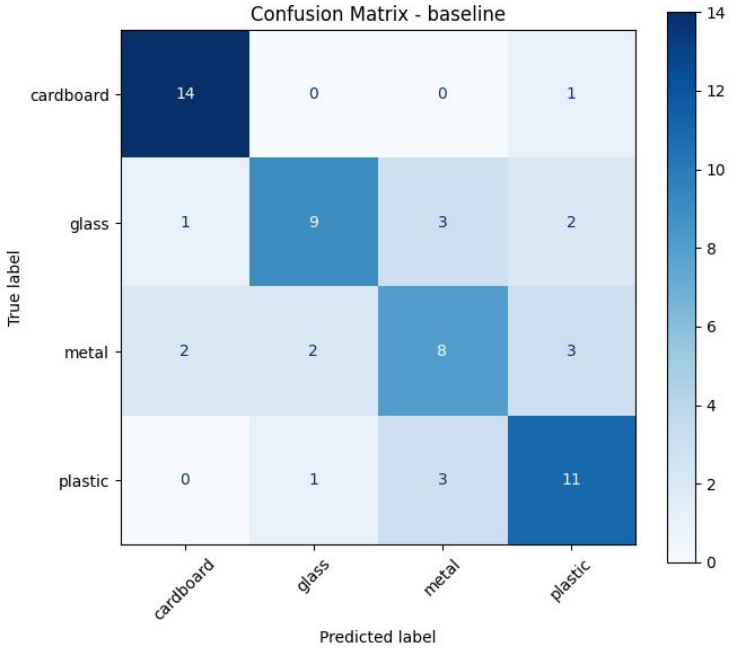
Interpretation

Synthetic data helped slightly. The gain is modest because the test set is small and the classes overlap visually.

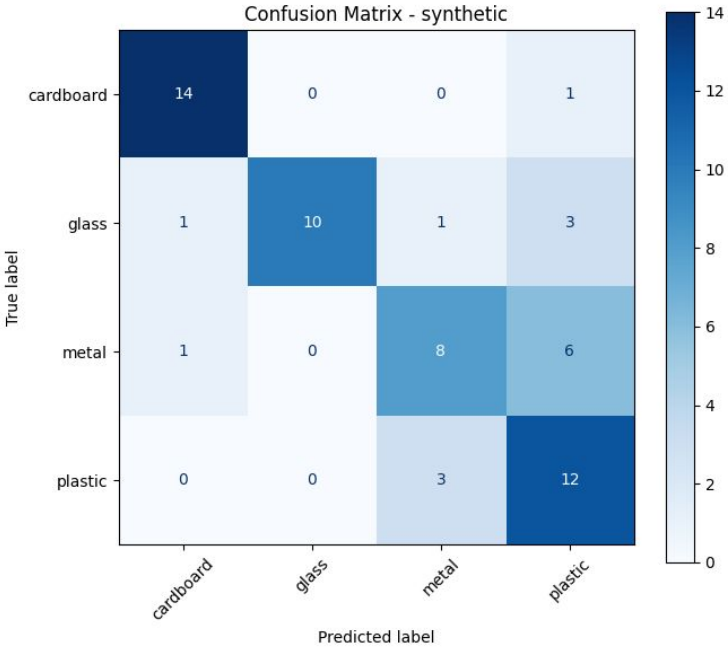
Reported from the test-set confusion matrices.

Confusion Matrix: Baseline vs Best Model

The best model improved some confusions, but material overlap remains.



Baseline: 70.00%



Synthetic: 73.33%

Takeaway: cardboard is easiest; glass, plastic, and metal are harder because they share shine, transparency, and packaging cues.

Error Analysis: What Failed and Why?

Misclassified examples reveal when the model relies on visual shortcuts instead of true material properties.

Glass → Cardboard

Broken glass was small, dirty, and placed on a pale background, so the model focused more on flat color/texture than transparent edges.

Plastic → Metal

Some plastic bottles had dark colors, glossy labels, and reflections that made them look metal-like.

Metal → Plastic

Crumpled foil and cans looked like flexible plastic packaging because of their shape, labels, and shiny surfaces.

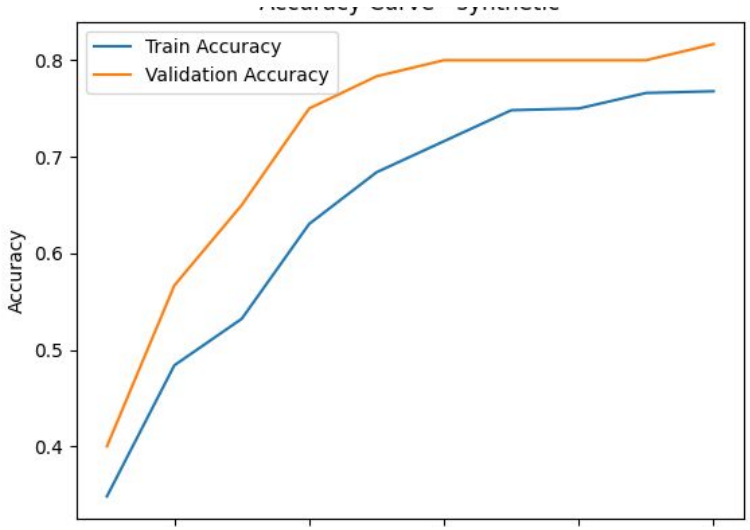
Cardboard → Plastic

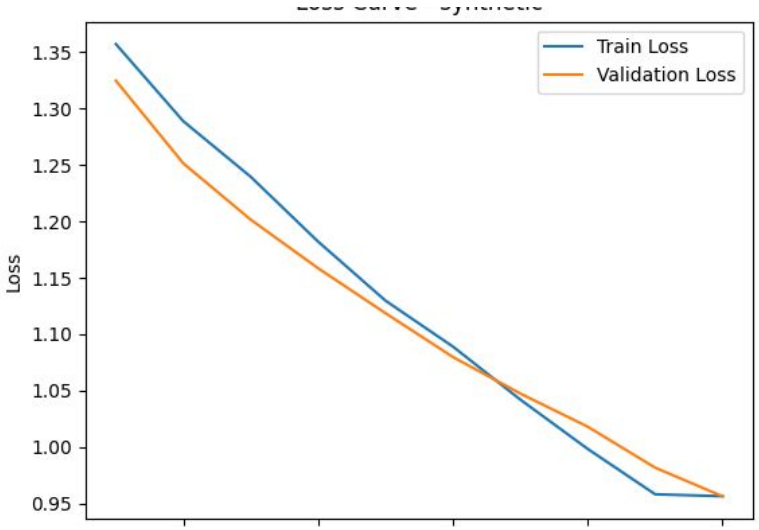
Printed cartons with bright packaging and straws looked more like plastic containers than plain cardboard.

Confidence Observation: Most wrong predictions had low confidence (0.30–0.48), indicating model uncertainty rather than definitive error.

Training Curves: Test Accuracy Decides

Notebook evidence shows learning was stable; validation performance alone was not enough.





Notebook insight

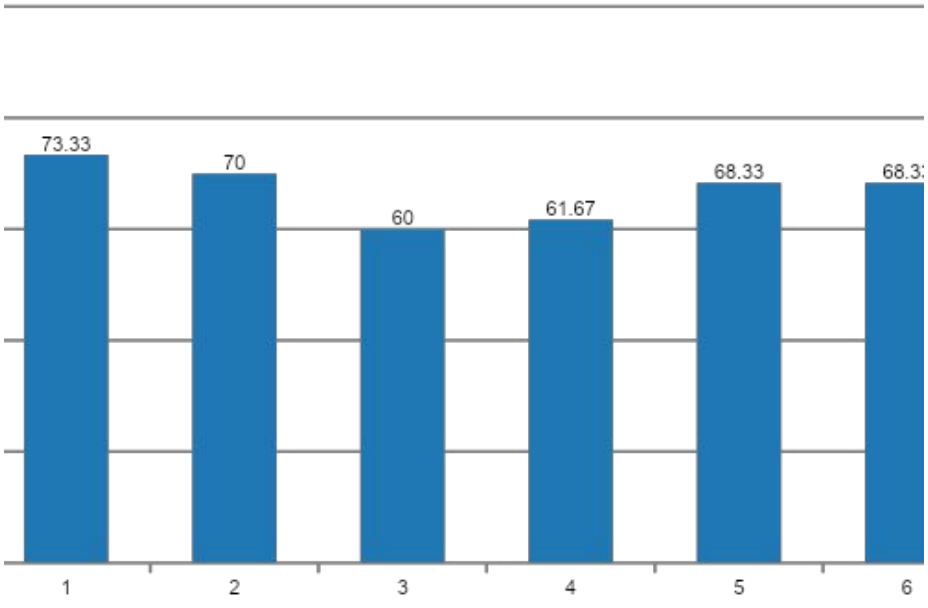
Synthetic-only achieved the best final test accuracy even though synthetic + augmented showed strong validation behavior. This suggests that more perturbation is not always better.

What we learned

The model was learning steadily, but the real test set exposed the final generalization gap between configurations.

Robustness: Best Model Under Real-World Image Changes

Evaluation of model performance across diverse environmental perturbations.



Smallest drop

Blur dropped accuracy from 73.33% to 70.00%.

Largest weaknesses

Noise and dark lighting caused the largest drops: 60.00% and 61.67%.

Interpretation

The model is somewhat robust to blur and occlusion, but pixel-level noise and low-light images still hurt material recognition.

Key Takeaway: While resilient to basic distortions, specialized augmentation for low-light and noise-heavy environments is necessary for field deployment.

Final Takeaways & Future Directions

Key Takeaway

Transfer learning works well and synthetic data provides a **3% accuracy boost (70% to 73%)**. However, the model still relies on visual shortcuts rather than true material identity.

Limitations

Struggles with visually similar materials (glass, plastic, metal) due to shared optical properties. Performance degrades significantly under noise and low-light conditions.

Originality & Contribution

Custom 4-class framing of RealWaste data, balanced splits, and a train-only synthetic pipeline. Innovation includes robustness test design and specific parameter/normalization tuning.

Future Improvements

- Fine-tune deeper layers & test backbones (ResNet-50)
- Generate more realistic synthetic datasets
- Implement confidence calibration for deployment

References & Resources:

https://docs.pytorch.org/vision/main/models/efficientnet.html
https://archive.ics.uci.edu/dataset/908/realwaste